Supplementary Step 6 Visualization Report

Overview

Metric	Value
Representative geometry session	PID001 / 2025-11-06_008
Included participant-sessions from Step 6	36
Included questions in Step 6 trial table	30
Abstract onset-locked epochs used	539
Concrete onset-locked epochs used	536
Epoch window	-3.5 s to 13.2 s around question onset
Baseline	-2.0 s to 0.0 s
Long-channel montage used in descriptive plots	22 pairs (44 chromophore channels)
Primary Step 6 adjusted p-value	0.052102

Adaptation note. This report reuses the same supplementary visualization family created for Step 5, but anchors it explicitly to the Step 6 trial-model table. Because Step 6 freezes the Step 5 cohort and outcome definition, the onset-locked descriptive plots shown here are based on the same included trial set that feeds the covariate-adjusted mixed-effects models.

Rendering note. As in the earlier supplementary visualization reports, the sensor-geometry figure uses the participant digitization point cloud plus source, detector, channel, and pair geometry rather than an fsaverage brain-surface rendering, because the local `pyvista/fsaverage` stack is not available in this environment.

Step 6 Main Figures Reused

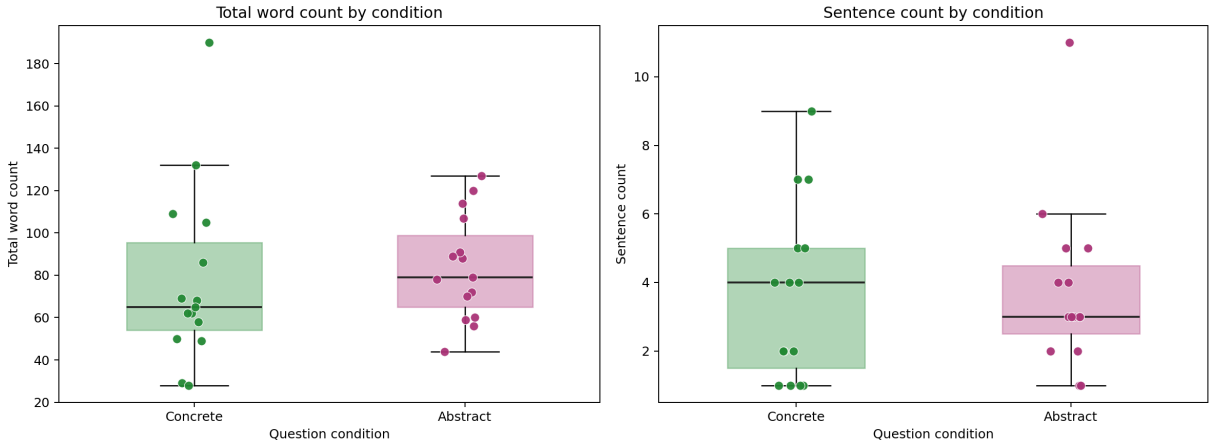

Figure 1: Step 6 item-level balance plots for total word count and sentence count.

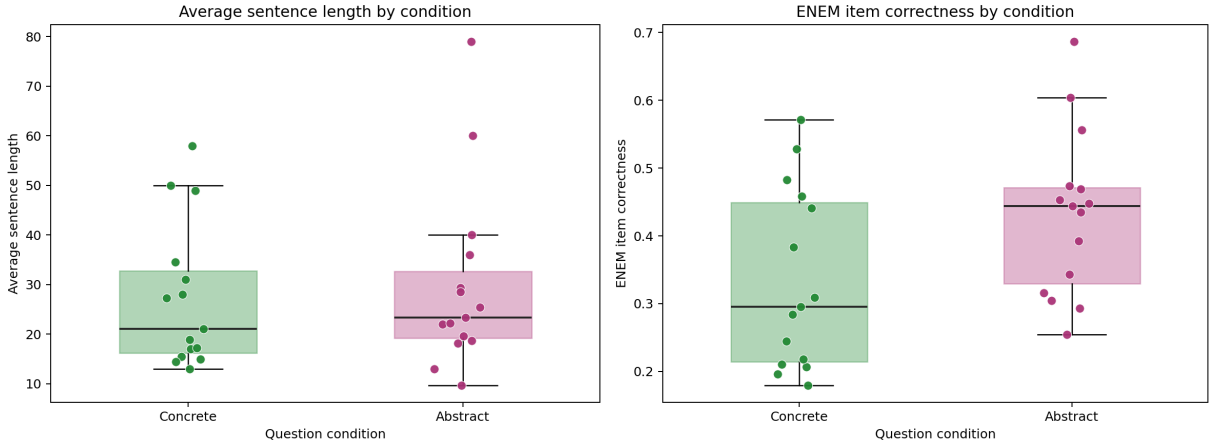

Figure 2: Step 6 item-level balance plots for sentence length and ENEM item correctness.

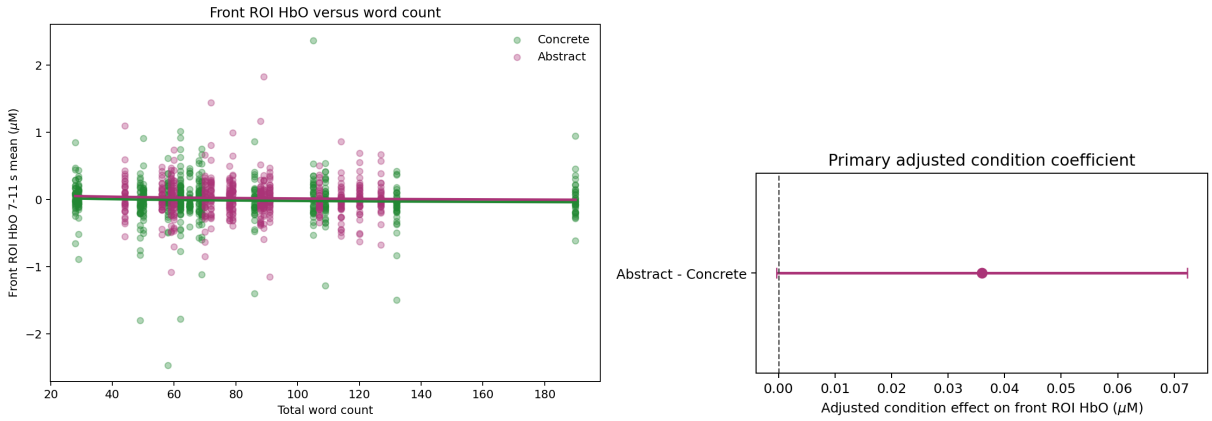

Figure 3: Step 6 scatter and partial-effect figures from the primary covariate-adjusted model.

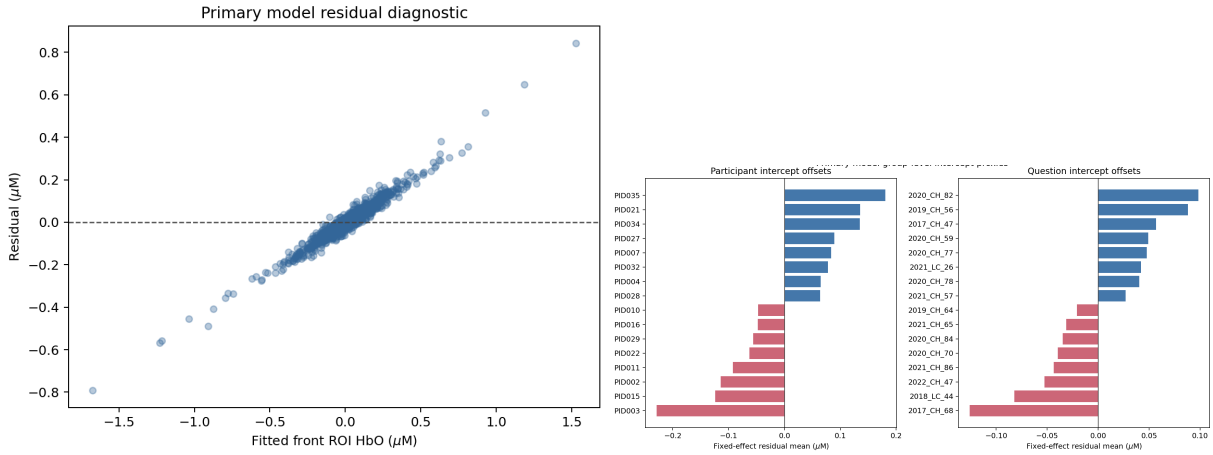

Figure 4: Step 6 residual diagnostic and group-level intercept-proxy plots from the primary model.

Step 5-Style Supplementary Plots for Step 6

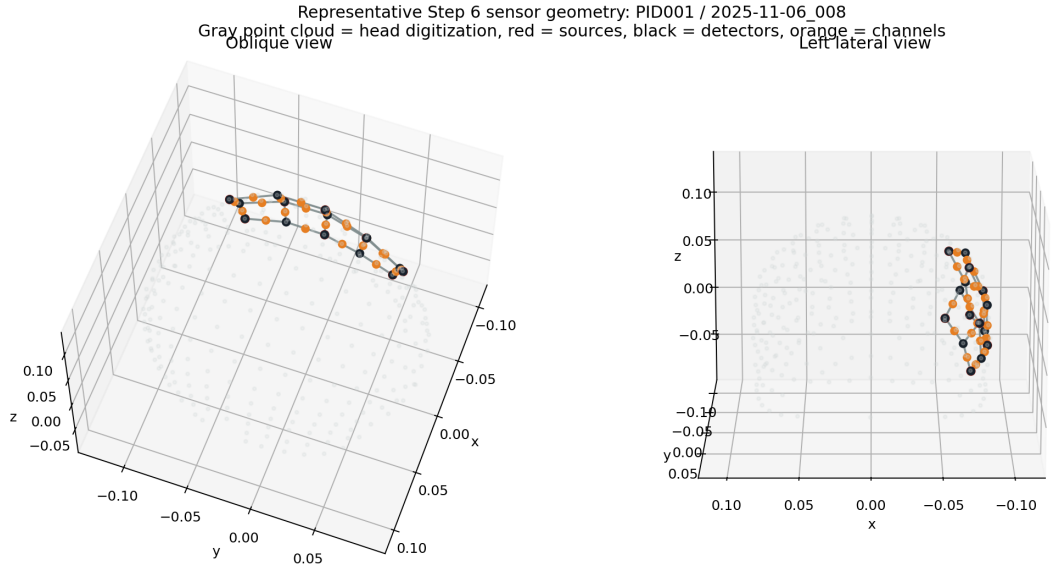

Figure 5: Representative optode, channel, and source-detector-pair geometry for one Step 6 included participant-session. Short pairs are shown with dashed orange lines.

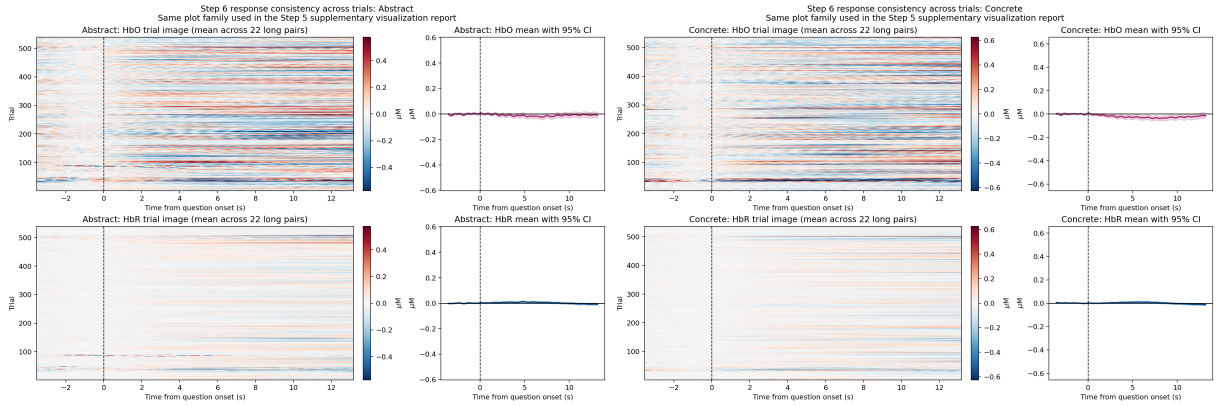

Figure 6: Condition-adapted trial-consistency image plots for the Step 6 trial set.

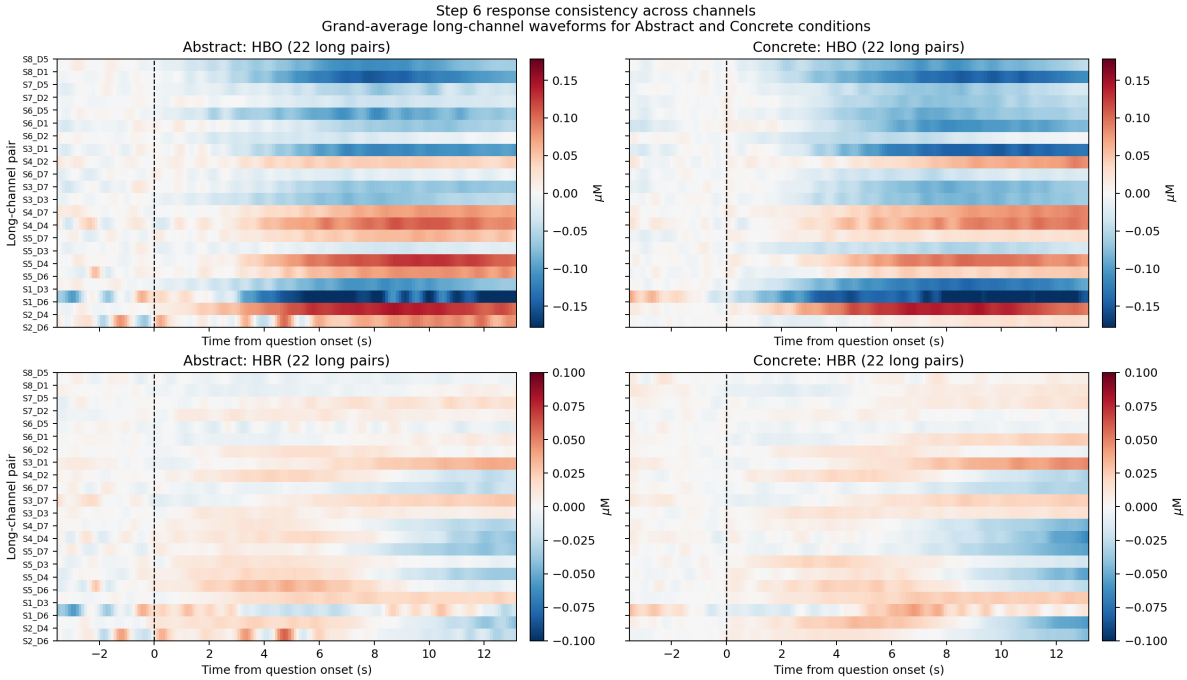

Figure 7: Grand-average long-channel response images across channels for the Abstract and Concrete conditions.

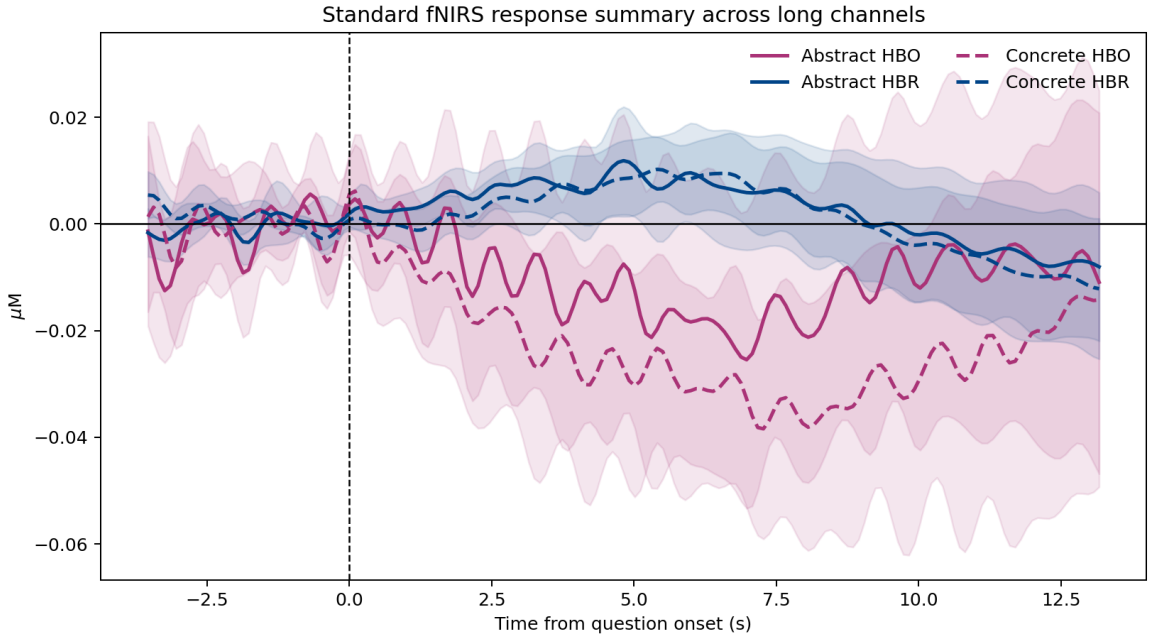

Figure 8: Standard fNIRS waveform comparison across long channels for the Step 6 trial set.

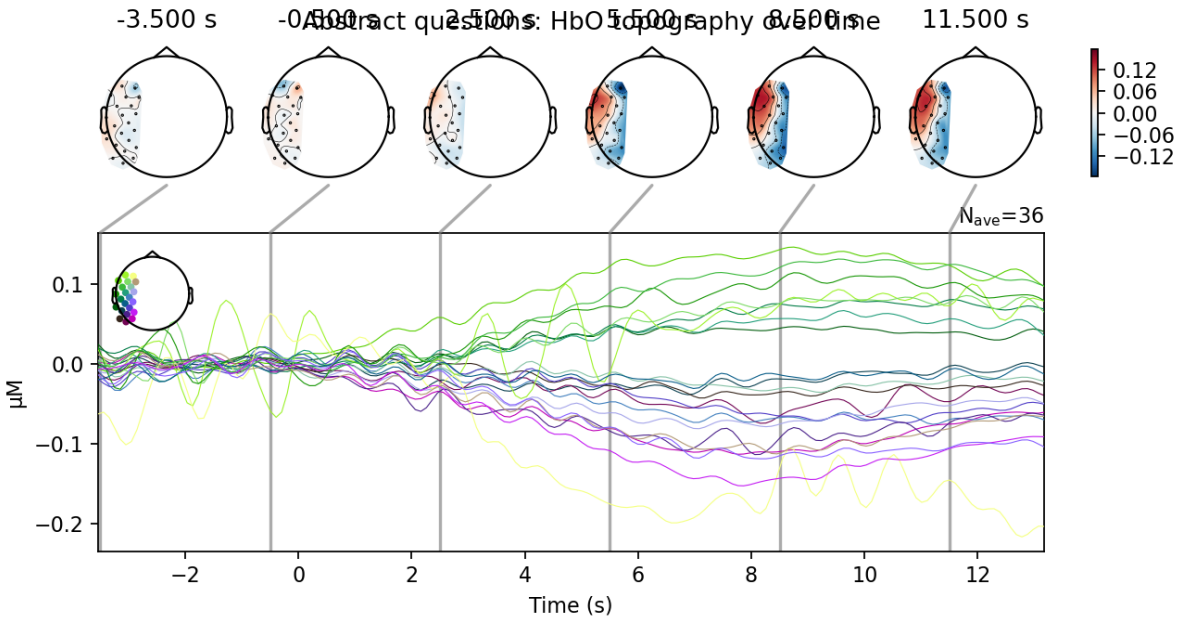

Figure 9: Grand-average HbO topography over time for Abstract questions using the Step 6 included trial set.

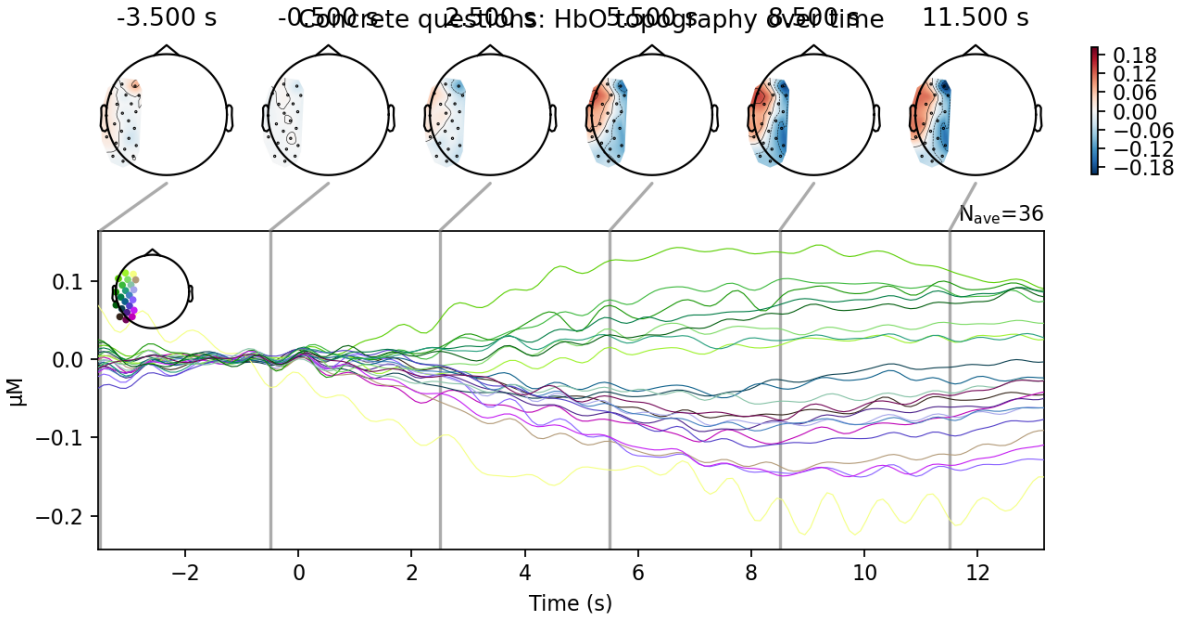

Figure 10: Grand-average HbO topography over time for Concrete questions using the Step 6 included trial set.

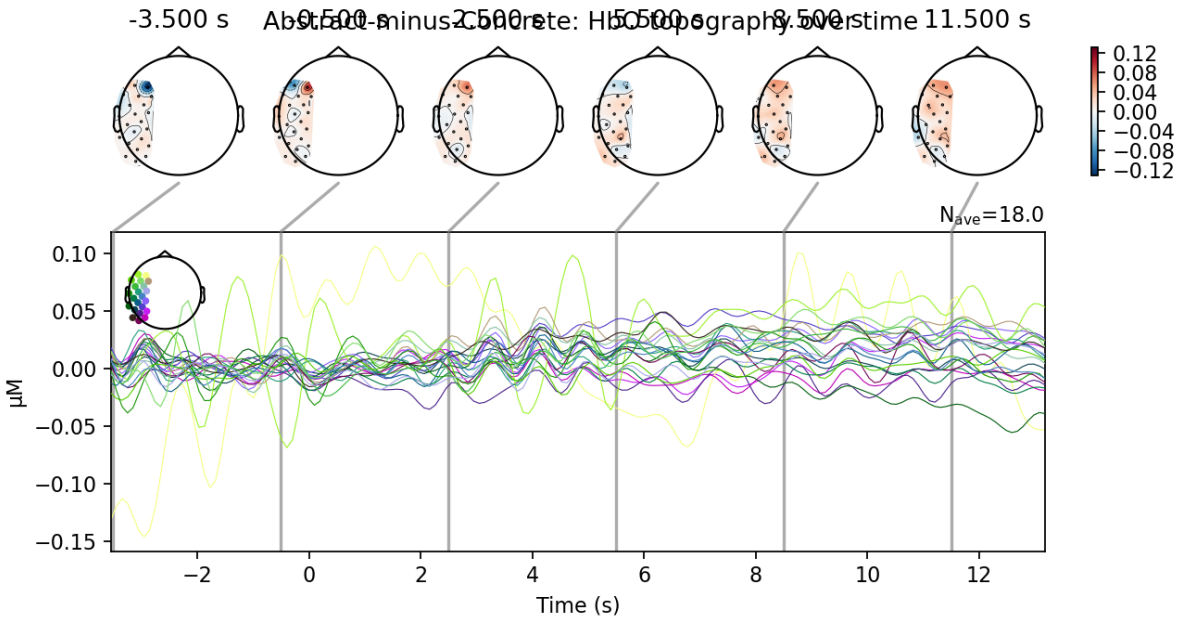

Figure 11: Grand-average Abstract-minus-Concrete HbO topography over time.

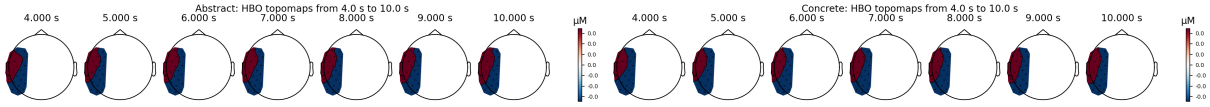

Figure 12: HbO topographic time series from 4.0 s to 10.0 s for the Abstract and Concrete conditions.

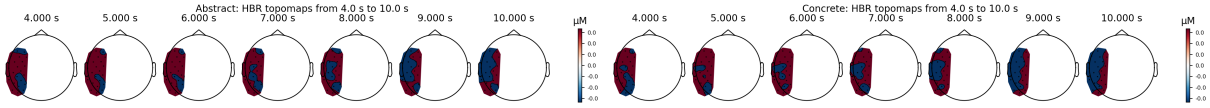

Figure 13: HbR topographic time series from 4.0 s to 10.0 s for the Abstract and Concrete conditions.

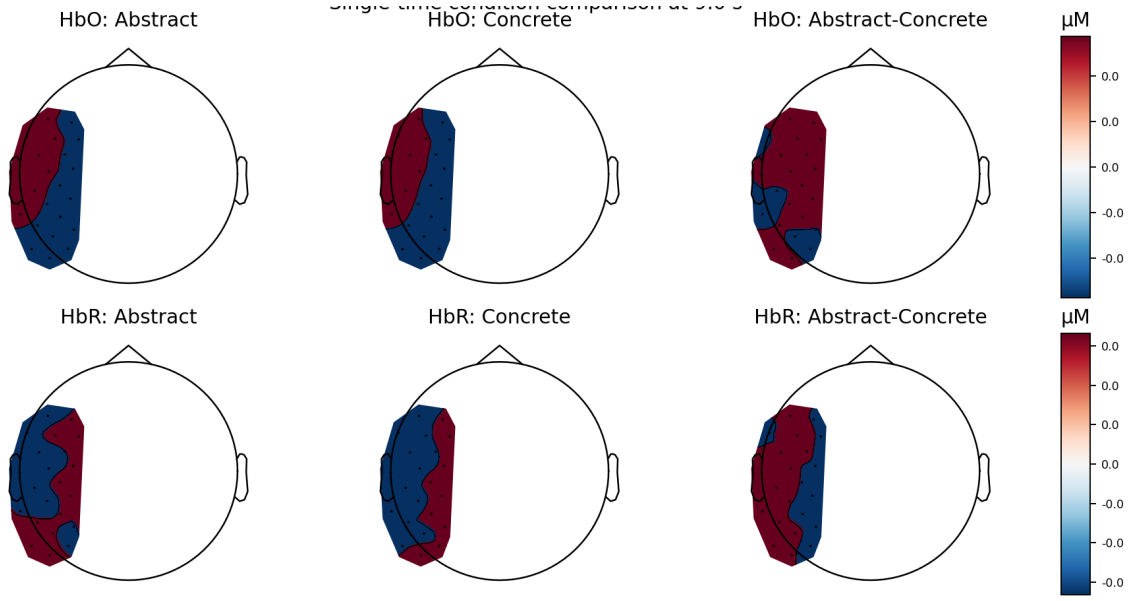

Figure 14: Single-time comparison at 9.0 s for Abstract, Concrete, and Abstract-minus-Concrete topographies for both HbO and HbR.

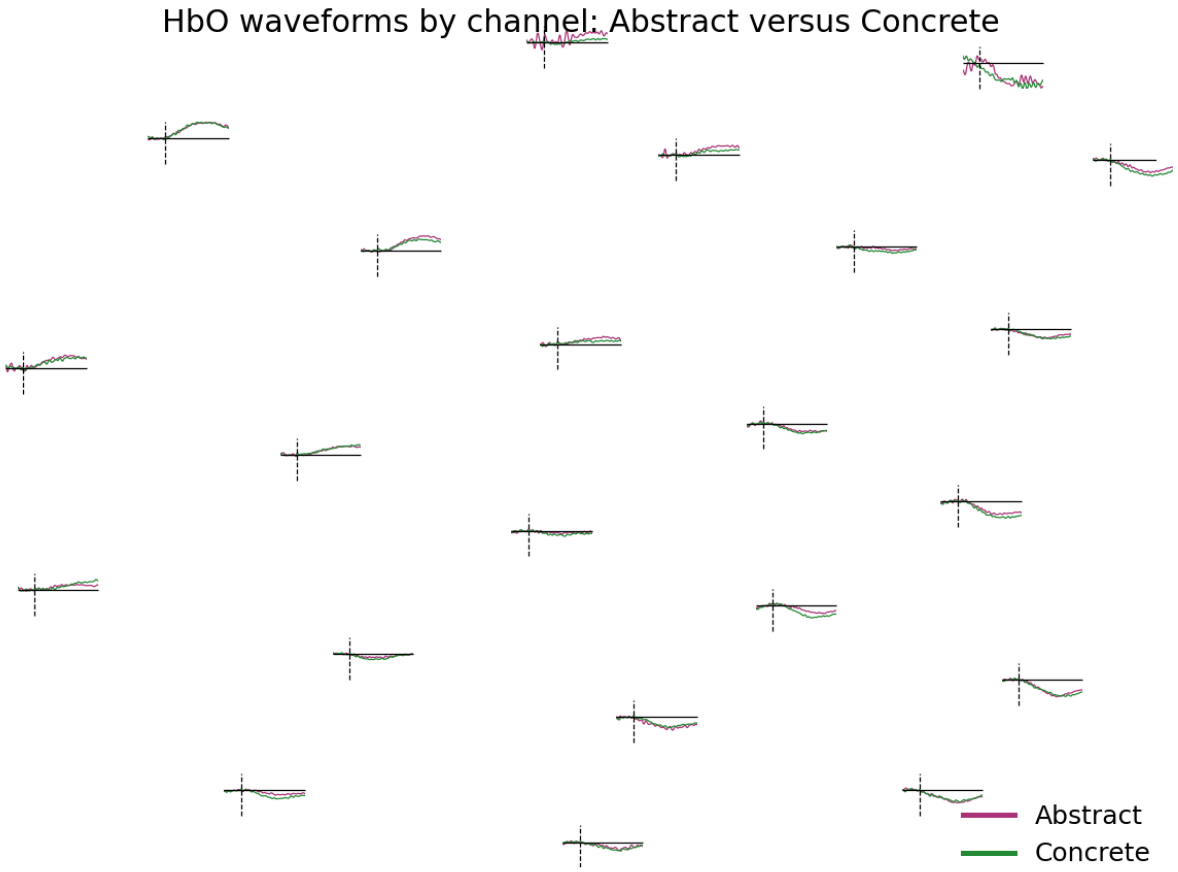

Figure 15: HbO waveforms by long channel, overlaid for Abstract and Concrete, to show what drives the descriptive topographic differences.